

Geospatial Analysis on Public Healthcare Facilities in Myanmar

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Course Name : Geospatial Analysis

Institution : GHEA21 Spring, 2026

Date : 18th May, 2026

Abstract

The report mainly describes the accessibility of healthcare facilities in Myanmar across the geographical distribution in Myanmar. The three main datasets used in this geospatial analysis report are the 2016 census population data in Myanmar, the 2020 health facilities vector data (including hospitals across 330 townships and bed capacity of the hospitals) and township boundaries vector data. The geospatial methods such as spatial joins, population and bed capacity ratio analysis, calculation of the nearest neighbor distance calculation and interactive mapping are used for in-depth analysis. Overall, among the 1,130 facilities recorded, rural townships in Shan states and Kayin have significant infrastructure disparity and a huge gap exists among the major cities, upper and lower regions of Myanmar. The findings in the report depict the insightful evaluation on the current healthcare facilities distribution in Myanmar covering all regions and states in Myanmar.

Introduction

Myanmar is one of the second largest countries in Southeast Asia and it has an overall population of 51 million people nationwide. Despite having strong historical and cultural backgrounds with 135 ethnicities, Myanmar has been facing political instability since 2021 which has led the country to struggle in different aspects of nation building (Myanmar Information Management Unit, 2026). Especially, related to the healthcare infrastructure, the global health security index of Myanmar is ranked 85 out of 195. This shows that significant lack of infrastructure and limited capacity expose the country's population to healthcare challenges (Nuclear Threat Initiative & Johns Hopkins Center for Health Security, 2021).

Therefore, this report is documentation as a systematic case study and analysis on the health infrastructure across the country. It predominantly emphasizes on the network of General Hospitals as the biggest facilities, township hospitals and station hospitals to be an evidence-based evaluation.

As for the official regions/states and union territory in Myanmar, it is identified as (15) altogether, which are (7) regions as Sagaing, Mandalay, Magway, Yangon, Taninthryi, Bago (East, West), Ayeyarwady and (7) states as Mon, Kachin, Kayah, Kayin, Chin, Rakhine, Shan (East, South, North) and union territory as Nay Pyi Taw.

Research Question

The main research question for this report is

“ How are health facilities in Myanmar allocated, according to data from 2020, and which region and township has the significant gap for accessibility of healthcare in terms of bed capacity per population ratio, spatial distribution and number of facilities? “

As a primary investigation for this main research question, the report examines the highest and lowest counts of healthcare facilities, distribution of hospital types and the proportion of the population size and bed capacity by both states/regions and township levels.

Data Sources and Description

There are three main datasets used in this report. The first dataset is the health facilities dataset which is from the Myanmar Information Management Unit Data. The second dataset is the census population from the Myanmar Information Management Census Baseline from 2016. The third one is the township boundaries dataset as GeoNode from Web Feature Service.

Table 1: Data Description

Index	Health Facilities Data Set	Population Dataset	Township Boundaries Dataset
Type	Vector Data (Point)	Tabular Data	Vector Data (Polygon)

Rows	1,130 rows	330 townships	330 polygons
Main Fields	Bed Class, TS_PCODE, Well-known Text coordinates(the_geom), Type of Facility, SR	TS_PCODE, Population for both sexes	TS, TS_PCODE, geometry, SR

Term : TS_PCODE = Place Code of Township

SR = State/Region

As geospatial data, the GPS coordinates as vector points from hospitals' maps and the vector polygons of the shapes for townships by using the web feature service are mainly synced with common key column "TS_PCODE".

Data Preprocessing Steps

As data preprocessing steps, the point coordinates from hospital data is converted from string to numeric coordinates. Plus, the "NA" strings in the bedClass column are also transformed to numeric values as NaaN. As the main data column, the text in PCODE columns are adjusted (like standardizing uppercase and removing whitespaces). The GeoNode of township boundaries are also modified with two coordinate systems, UTM Zone 47N (EPSG:32647) for calculation of the area and CRS: WGS84 (EPSG:4326) for location mapping. In addition to that, all of the data sources are merged using the key TS_PCODE.

Methodology

The analysis follows a certain workflow of geospatial methods as mentioned.

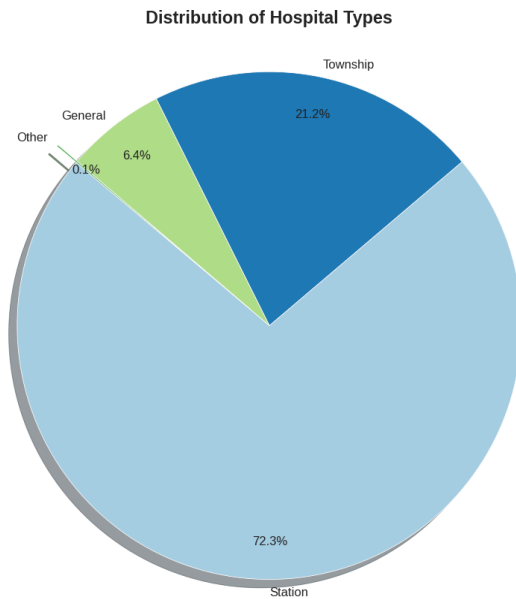
Table 2: Methodology

Geospatial Methods	Purpose	Formulation
Spatial Joins	The function integrates the health facility and township polygons with the main keys PCODE for State/Region and Townships to investigate the facility distribution.	geopandas merg()
Coordinate Reference System	This is used for mapping and calculation of area as (km ²) intended for further analysis with population and area.	Gdf.to_crs ('EPSG:32647)
Beds per population (x10,000) (World Health Organization, 2026)	The total beds are divided by the population for fair comparison between different regions and townships.	Bed_total / population * 10,000
Facility density per 100 km ²	The facility count is normalized with area covered for better	Facility count/ km ² of the area * 100

	understanding of accessibility.	
Nearest-Neighbour Distance (cKDTree)	It uses the straight-line method for calculating the distance from the center of the township to the nearest health facility.	scipy.spatial.cKDTree
Choropleth Mapping	It is used for visualization of both static and interactive maps of the geospatial analysis.	folium Choropleth/ geopandas plot ()
Folium and Marker Cluster Web map	It is used to eliminate overplotting and visualize interactive Choropleth Map and MarkerCluster map.	folium.MarkerCluser()

Results and Discussion

Exploratory Analysis



Hospital Types Distribution in Myanmar (2020)

Hospital types:

lvlHsp_eng

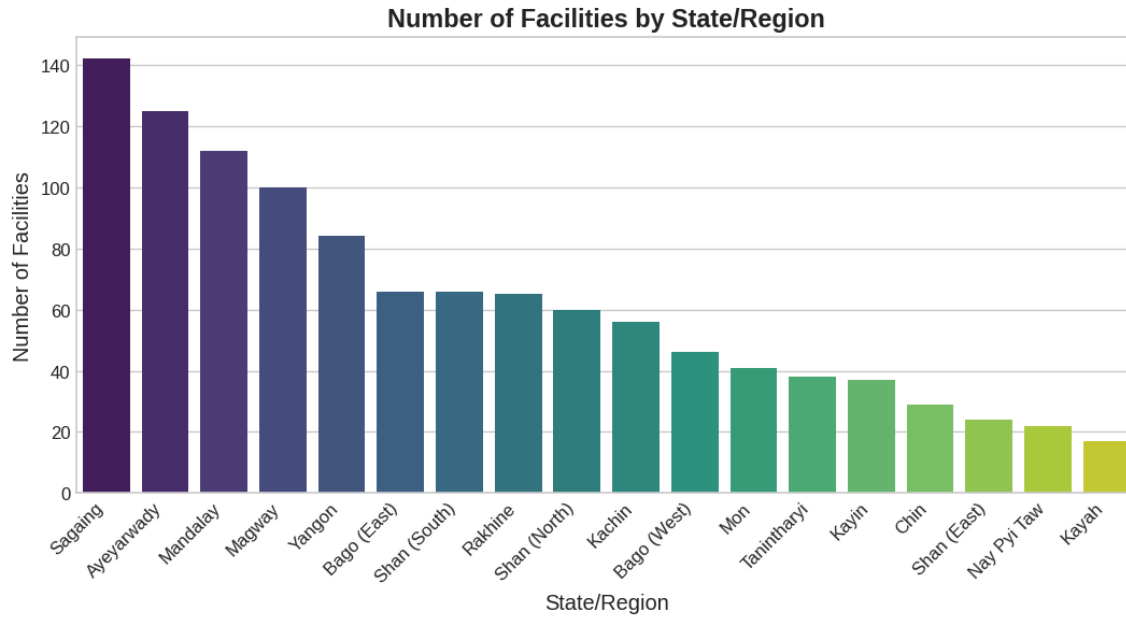
Station 769

Township 226

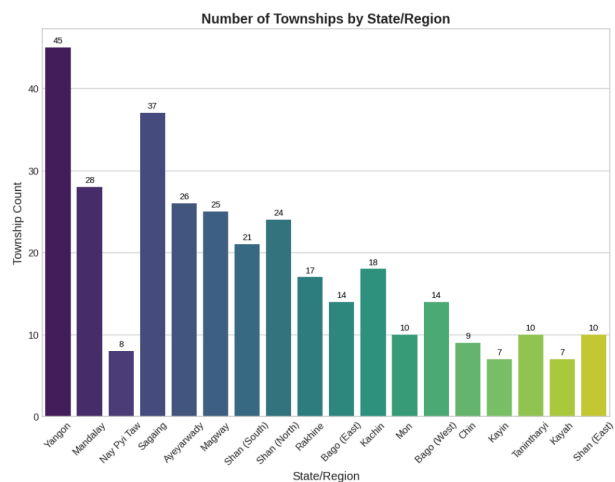
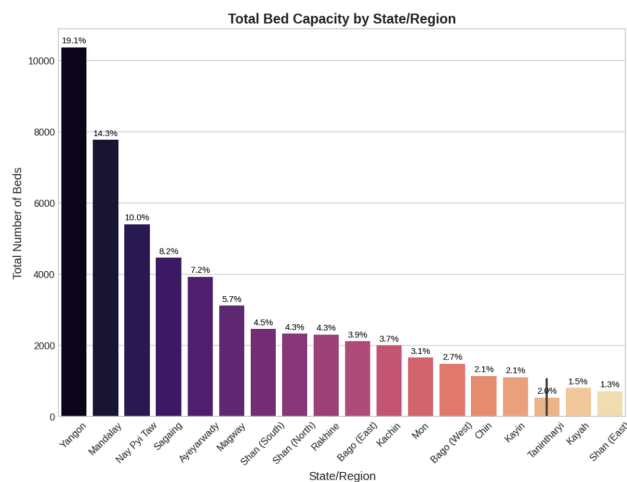
General 68

District 1

Among the types of hospitals in Myanmar, station level hospitals are the highest with 72.33%, followed by the township hospitals (21.2%), General Hospitals (6.4%) and others as district hospitals (0.1%).

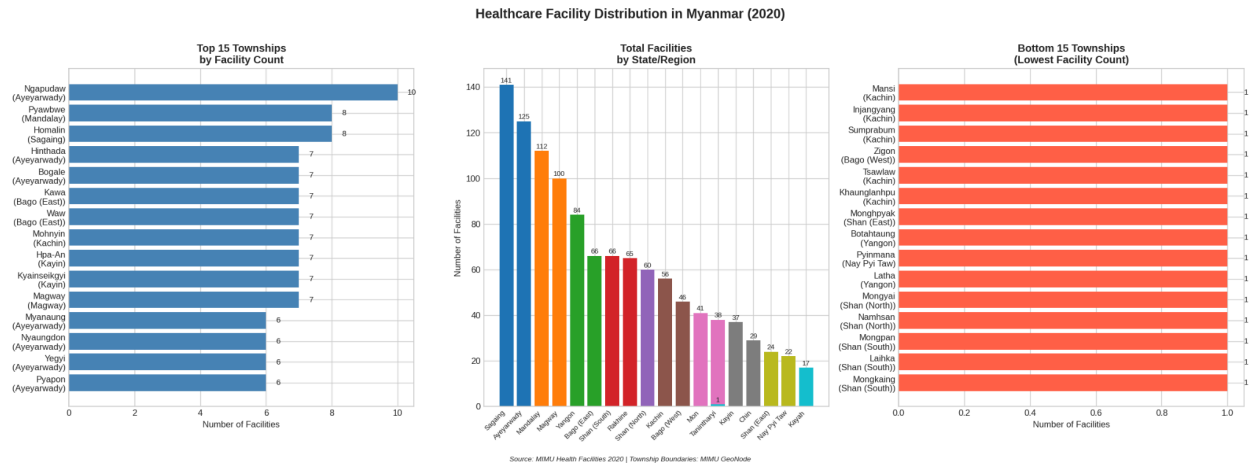


According to classification with state/region (SR), the top three regions are Sagaing, Ayeyarwady and Mandalay which are more than 100 of public hospitals for each region. This can be because these regions are included in the largest regions of Myanmar in area compared to others. Moreover, Kayah and Nay Pyi Taw are analysed as the lowest facility, being the smallest regions nationwide.



The chart highlights the total bed capacity by regions (percentage), which shows that Tanintharyi (2%), Kayah (1.5%) and Shan-East (1.3%) are the lowest. As for the analysis of the number of townships, Yangon,Sagaing, Mandalay, Ayeyearwaday, Magway are the highest, and Kayin, Kayah and Nay Pyi Taw have the lowest number of townships.

ANALYSIS 1: FACILITY COUNT RANKINGS

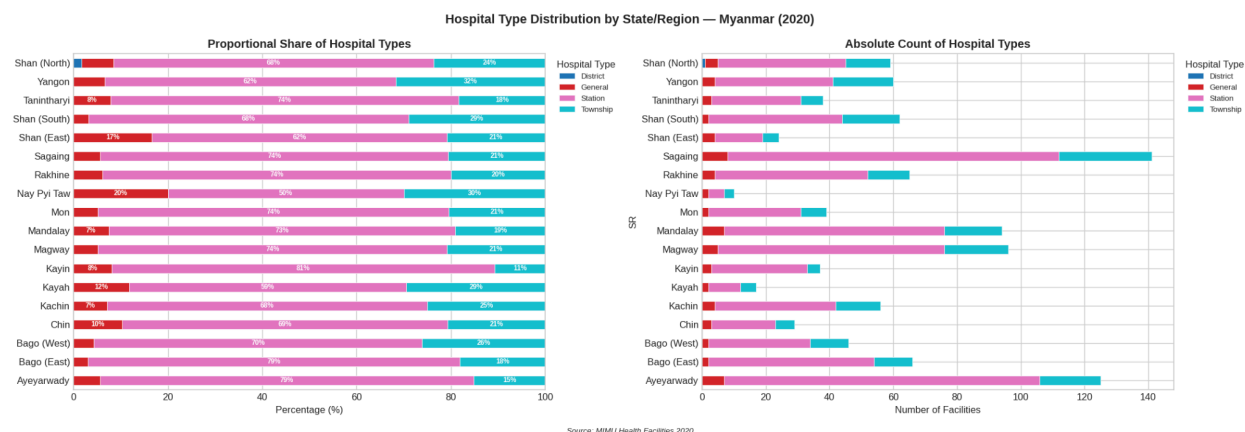


State/Region Summary:

SR_EN	facility_count	bed_total	num_townships
Sagaing	141	4458.0	37
Ayeyarwady	125	3921.0	26
Mandalay	112	7762.0	28
Magway	100	3115.0	25
Yangon	84	10363.0	45
Bago (East)	66	2116.0	14
Shan (South)	66	2458.0	21
Rakhine	65	2304.0	17
Shan (North)	60	2326.0	24
Kachin	56	1998.0	18
Bago (West)	46	1480.0	14
Mon	41	1655.0	10
Tanintharyi	38	1061.0	10
Kayin	37	1111.0	7
Chin	29	1145.0	9
Shan (East)	24	717.0	10
Nay Pyi Taw	22	5398.0	8
Kayah	17	803.0	7
Tanintharyi	1	0.0	10

The analysis depicts the top 15 townships having highest facility counts, and the bottom 15 townships which have the lowest public hospitals. As for the bed capacity counts based on the regions/states, it is examined that facility count and the bed capacity are not directly proportional. Significantly, Yangon has the highest bed capacity with 10147 in numbers, followed by Mandalay (7762 bed counts) and Nay Pyi Taw (5398 bed counts). Based on the urbanness and development criteria, Yangon(major commercially important city), Mandalay and Nay Pyi Taw (capital city) are in the top ranks nationwide. Therefore, it can be stated that the major regions in Myanmar have the highest bed capacity as healthcare facilities compared to other regions. As a notable finding, Nay Pyi Taw (union territory) has the outlier effect, even being in one of the least facility count, the bed capacity has been identified as third highest which can be due to the large specialized hospitals (general hospitals).

ANALYSIS 2: HOSPITAL TYPE PROPORTIONS



Proportions table (%):

lvlHsp_eng	District	General	Station	Township
SR				
Ayeyarwady	0.0	5.6	79.2	15.2
Bago (East)	0.0	3.0	78.8	18.2
Bago (West)	0.0	4.3	69.6	26.1
Chin	0.0	10.3	69.0	20.7

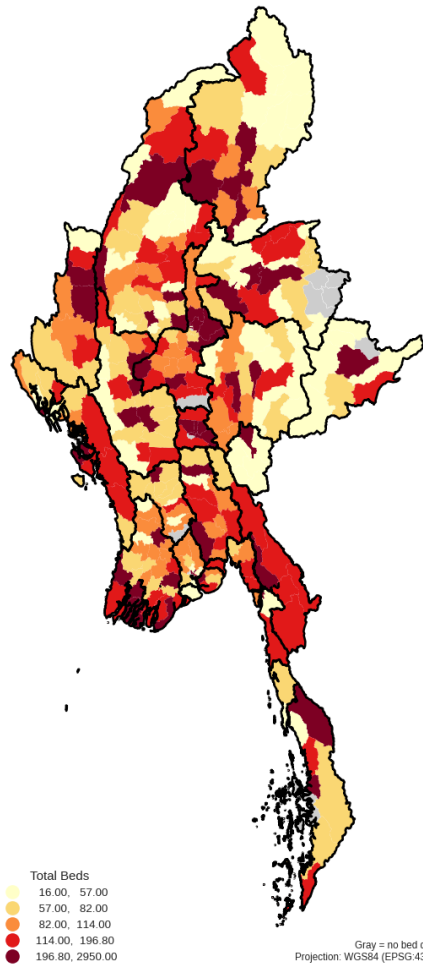
Kachin	0.0	7.1	67.9	25.0
Kayah	0.0	11.8	58.8	29.4
Kayin	0.0	8.1	81.1	10.8
Magway	0.0	5.2	74.0	20.8
Mandalay	0.0	7.4	73.4	19.1
Mon	0.0	5.1	74.4	20.5
Nay Pyi Taw	0.0	20.0	50.0	30.0
Rakhine	0.0	6.2	73.8	20.0
Sagaing	0.0	5.7	73.8	20.6
Shan (East)	0.0	16.7	62.5	20.8
Shan (South)	0.0	3.2	67.7	29.0
Tanintharyi	0.0	7.9	73.7	18.4
Yangon	0.0	6.7	61.7	31.7
Shan (North)	1.7	6.8	67.8	23.7

Based on the distribution of hospitals according to regions, the station hospitals are the highest in number of proportion compared to township hospitals. Nay Pyi Taw, Shan (East) and Kayah have the highest percentage of general hospitals, and Bago (East) and Shan (South) have the lowest percentage of general hospitals. For station hospitals, Kayin and Ayeyearwaday have the highest proportion, and Nay pyi Taw has a significantly low percentage of station hospitals. As for the township level hospitals proportion, Yangon is the highest which can be due to the fact that Yangon has the largest number of townships (31.7%), followed by Nay Pyi Taw (30%) and Kayah (29.4%).

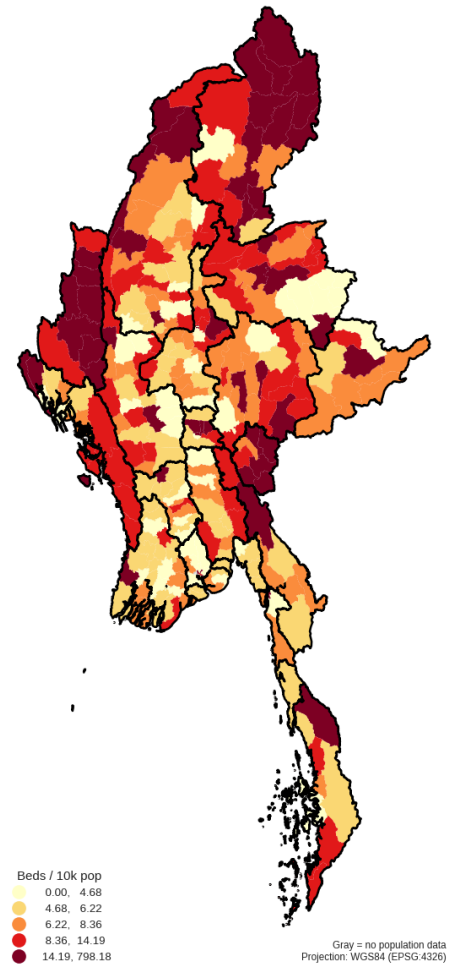
ANALYSIS 3: BED CAPACITY

Hospital Bed Capacity in Myanmar (2020)

Total Bed Capacity by Township



Beds per 10,000 Population by Township (WHO Standard Metric)



Top 10 townships by total beds:

TS_EN	SR_EN	bed_total	bed_mean
Chanayethazan	Mandalay	2950.0	590.0
Oke Ta Ra Thi Ri	Nay Pyi Taw	2050.0	410.0
Latha	Yangon	2000.0	2000.0
Lanmadaw	Yangon	1500.0	500.0
Zay Yar Thi Ri	Nay Pyi Taw	1375.0	343.8
Za Bu Thi Ri	Nay Pyi Taw	1050.0	525.0
Chanmyathazi	Mandalay	1000.0	500.0
North Okkalapa	Yangon	1000.0	500.0

Insein	Yangon	900.0	225.0
Taunggyi Shan	(South)	773.0	128.8

Top 10 townships by beds per 10k population:

TS_EN	SR_EN	beds_per_10k	Population
Latha	Yangon	798.2	25057.0
Lanmadaw	Yangon	318.1	47160.0
Oke Ta Ra Thi Ri	Nay Pyi Taw	251.2	81620.0
Dagon	Yangon	199.3	25082.0
Sumprabum	Kachin	196.4	2546.0
Chanayethazan	Mandalay	149.6	197175.0
Injyangyang	Kachin	144.3	1732.0
Cocokyun	Yangon	128.9	1940.0
Zay Yar Thi Ri	Nay Pyi Taw	123.5	111293.0
Za Bu Thi Ri	Nay Pyi Taw	95.1	110459.0

The first map shows the bed capacity per township level. Based on calculation, the top three townships of having highest bed totals are Chanyethazin township from Mandalay (2,950 beds), Oke Ta Ra Thi Ri township in Nay Pyi Taw (2,050 beds) and Latha township in Yangon (2,000).

With consideration on the population across the townships, beds per 10,000 population is illustrated in the second map. This is aimed to investigate which townships and areas have enough bed capacity based on the current population to make a more realistic conclusion. Based on the calculation, Latha township, Lanmadaw township and Oke Ta Ra Thi Ri have the highest rate of beds per 10K population.

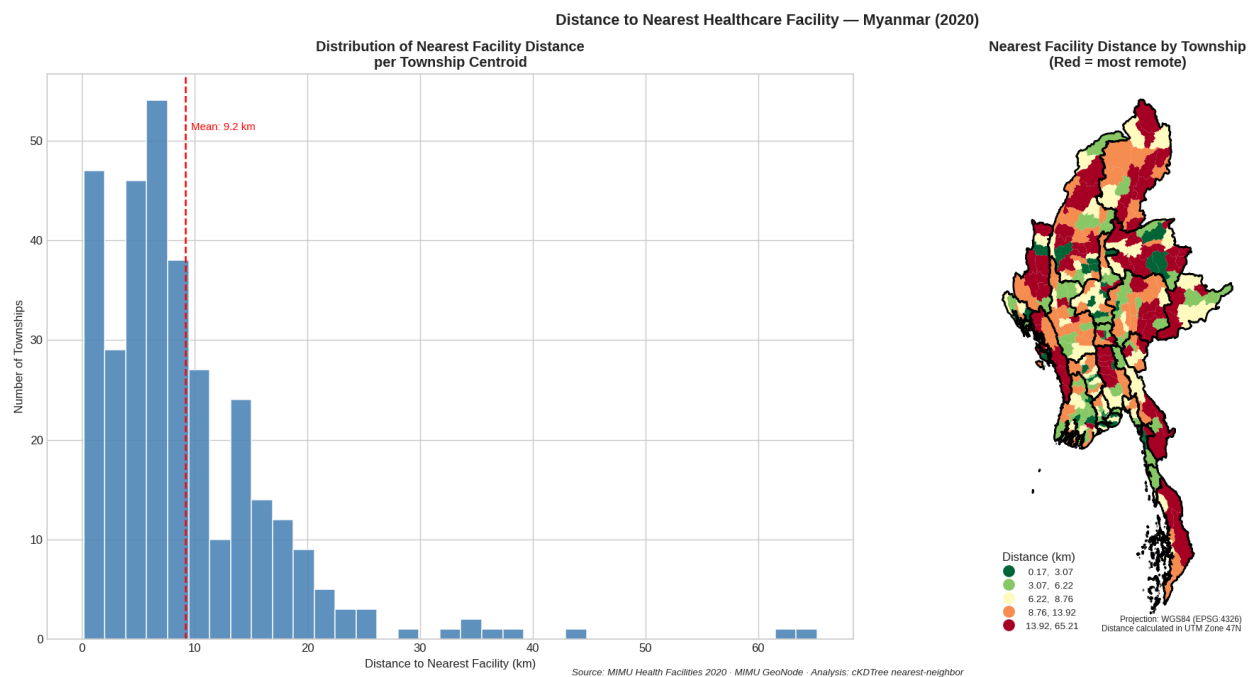
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ANALYSIS 4: NEAREST FACILITY DISTANCE
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Nearest distance stats (km):
count      330.00
mean        9.19
std         8.16
min         0.17
25%         4.17
50%         7.21
75%        11.98
max         65.21
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Townships >20 km from any facility: 23

Townships >50 km from any facility: 2

Top 10 most remote townships:

	TS_EN	SR_EN	nearest_km
	Narphan	Shan (North)	65.21
	Pangwaun	Shan (North)	62.63
	Pangsang (Panghkam)	Shan (North)	44.61
	Dawei	Tanintharyi	37.38
	Tsawlaw	Kachin	36.54
	Mongmao	Shan (North)	34.62
	Injangyang	Kachin	34.56
	Mongla	Shan (East)	33.06
	Cocokyun	Yangon	29.48
	Hakha	Chin	26.05



Since distance from the nearest hospital is also an important parameter to evaluate the healthcare accessibility, the nearest facility distance is plot. It is investigated that the mean distance to the near facility is 9.2km in each township.

In the map, the red area shows the most remote area, having a significant long distance from the nearest health facility. Some of the townships in Shan North, Shan East, Kachin

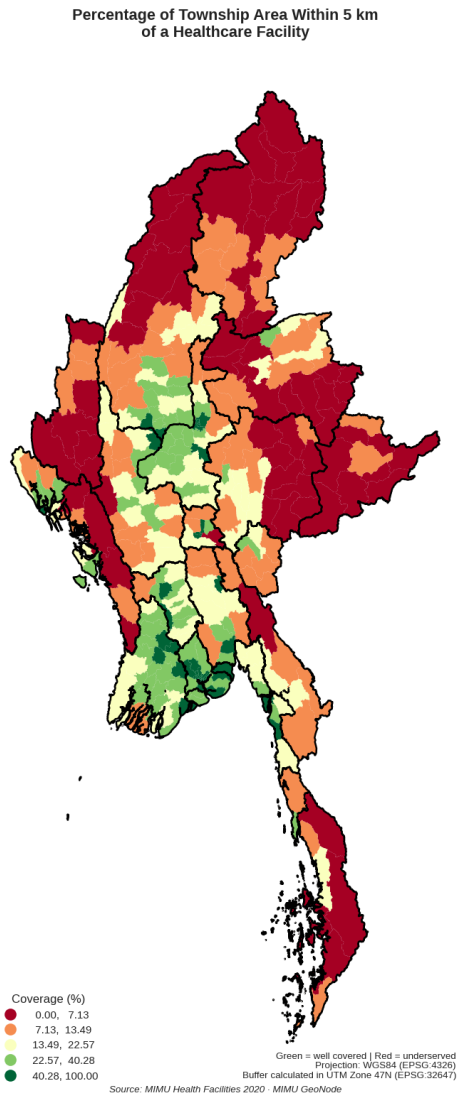
and Chin are identified as the most remote. According to the regional situation, these regions are underdeveloped compared to the large cities in Myanmar, which can be stated that the finding is valid. Even though Cocokyun from Yangon is also identified as most remote, it is the island with no residents, which can be neglected.

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ANALYSIS 5: 5 KM BUFFER COVERAGE

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Townships with 0% coverage: 5
Townships with <25% coverage: 215
Townships with >75% coverage: 38
National mean coverage: 28.0%



To have a deeper spatial analysis between township area and facility, the buffer coverage is calculated to identify the number of township areas within a 5km distance coverage rate. According to the analysis, there are 5 townships with 0% coverage area, and 215 townships with less than 25% area coverage within 5 km. Only 38 townships have the buffer coverage more than 75%. Therefore, it can be concluded that there is a huge gap between townships accessibility of facilities within the shortest possible period. Based on the map, Only the middle area of Myanmar has high coverage compared to the upper and lower part of Myanmar.

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ANALYSIS 6: HEALTHCARE ACCESSIBILITY INDEX (COMPOSITE)
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Top 10 townships (most accessible):

TS_EN	SR_EN	access_index	access_tier
Latha	Yangon	1.000	Very High
Lanmadaw	Yangon	0.781	Very High
Chanayethazan	Mandalay	0.457	Very High
Dagon	Yangon	0.432	Very High
Tamwe	Yangon	0.398	Very High
Botahtaung	Yangon	0.391	Very High
Kyeemyindaing	Yangon	0.382	Very High
Yankin	Yangon	0.378	Very High
Za Bu Thi Ri Nay Pyi Taw		0.350	Very High
Insein	Yangon	0.341	Very High

Bottom 10 townships (least accessible):

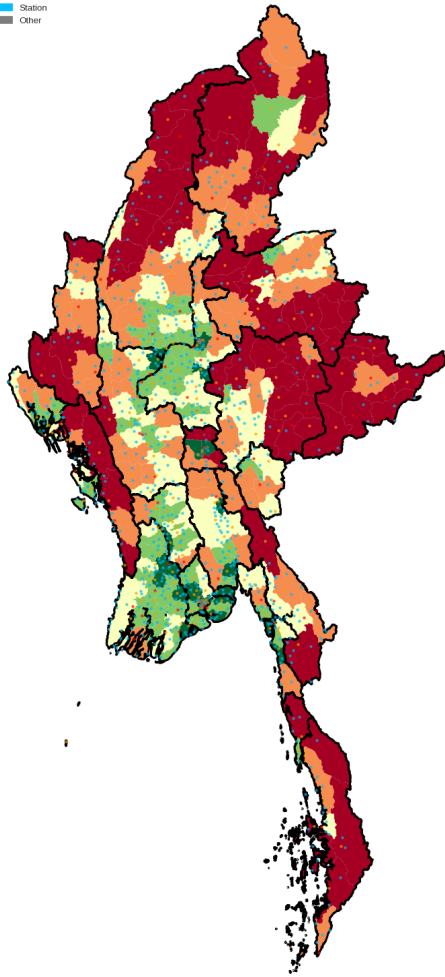
TS_EN	SR_EN	access_index	access_tier
Mongla	Shan (East)	0.000	Very Low
Mongmao	Shan (North)	0.000	Very Low
Pangwaun	Shan (North)	0.000	Very Low
Narphan	Shan (North)	0.000	Very Low
Pangsang (Panghkam)	Shan (North)	0.000	Very Low
Mongkaing	Shan (South)	0.008	Very Low
Mongping	Shan (East)	0.009	Very Low
Tanai	Kachin	0.009	Very Low
Tanintharyi	Tanintharyi	0.009	Very Low
Laihka	Shan (South)	0.011	Very Low

Mean Accessibility Index by State/Region:

SR_EN	
Yangon	0.290
Nay Pyi Taw	0.136
Mandalay	0.132
Mon	0.107
Ayeyarwady	0.092
Bago (West)	0.082
Bago (East)	0.080
Rakhine	0.059
Sagaing	0.059
Magway	0.058
Kayah	0.052
Shan (South)	0.038
Kayin	0.036
Chin	0.034
Tanintharyi	0.034
Kachin	0.033
Shan (North)	0.031
Shan (East)	0.018

Healthcare Accessibility Index by Township
(Composite Score: Facility Density + Bed Capacity + 5km Coverage)

Facility Type
■ General
■ Township
■ Station
■ Other



Index = 35% Facility Density + 35% Beds/10k Pop + 30% 5km Coverage
 Projection: WGS84 (EPSG:4326)
 Source: MIMU Health Facilities 2020 - MIMU Census 2016 - MIMU GeoNode

Another level of in-depth analysis also performed by integrating the key three parameters (facility density as number of facilities with the area of 100 km², bed capacity per 10,000 population, and spatial coverage of facility within 5km buffer). The MinMaxScaler is used to scale within 0 to 1 with composition of facility density (35%), bed capacity (35%) and spatial coverage within buffer 5km (30%). According to the calculation and investigation on map, Yangon region has the highest accessibility index with (0.29), followed by Nay Pyi Taw(0.136) and Mandalay (0.132). Shan (East), Shan (North) and Kachin which are the upper regions of Myanmar have the lowest accessibility index. As overall evaluation, even

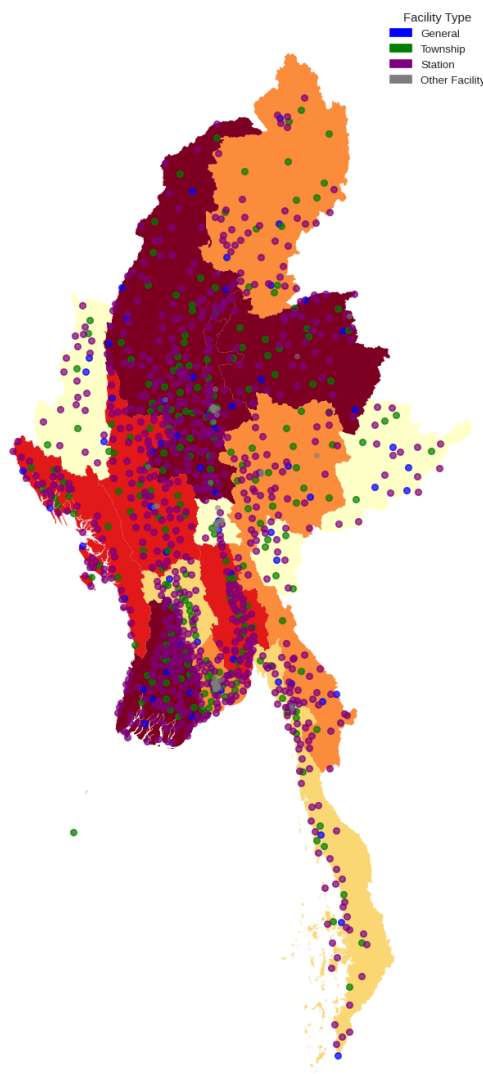
the highest healthcare accessibility index is 0.29 out of 1 which should have critical attention. The health facility in Myanmar needs more proactive initiatives such as having mobile health facilities, distributing the online health consultation by the government, and delivering more health infrastructure as well as manpower in the public health sector.

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PLOTTING NATIONAL HEALTHCARE RISK EXPOSURE

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National Healthcare Risk Exposure by State/Region with Facility Distribution



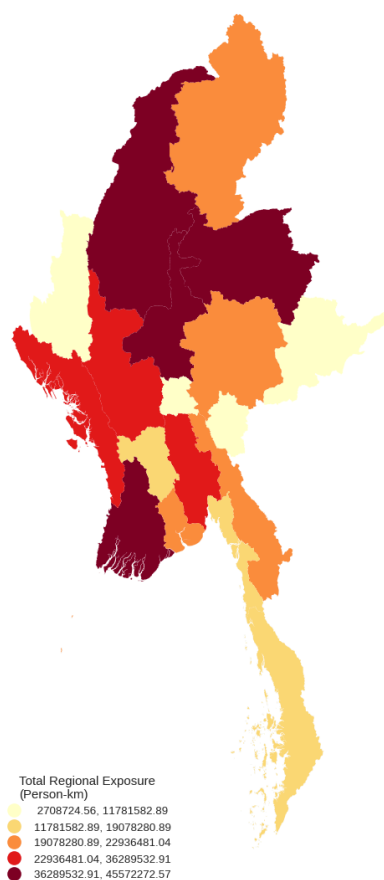
Source: MIMU Health Facilities 2020 - MIMU Census 2016 - MIMU GeoNode

The national healthcare risk exposure map clearly shows which regions have the highest risk, in which the massive population live away from the nearest hospital. This is first calculated by multiplying the parameter of population in each township with the nearest kilometer as distance to the nearest hospital. The, it sums the values of each township as respective state/region for total regional exposure. The dark red areas are the most risky areas which are Sagaing Region , Shan (North) State, Mandalay Region , Magway Region and Ayeyarwaddy Region.

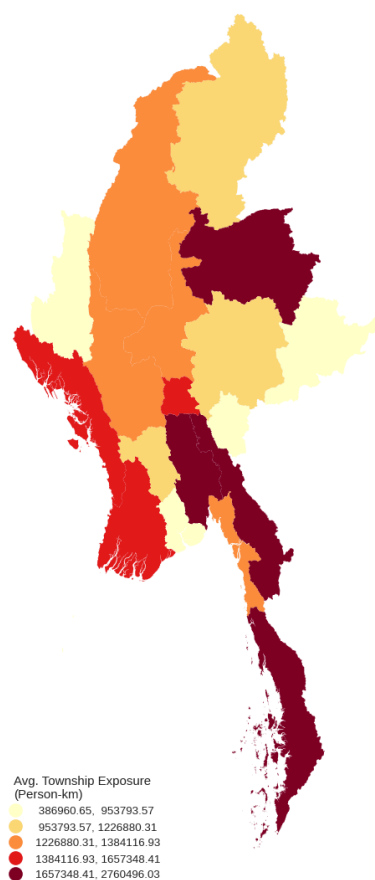
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PLOTING REGIONAL HEALTHCARE RISK (DUAL VIEW)
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Regional Healthcare Risk Exposure in Myanmar (2020)

Total Regional Exposure by State/Region



Average Township Exposure by State/Region



Source: MIMU Health Facilities 2020 - MIMU Census 2016 - MIMU GeoNode

According to the left map, it shows the potential risk of each region indicating where a large population lives a longer distance from hospitals. For the right map, it is calculated total regional exposure divided by the number of townships in the region or state. It gives insight about the underserved region, illustrating the typical level of risk per region identifying which region has a systemic struggle of distance to care. According to the map, the Th Mon State, Kayin State, Bago (East) Region, Tanintharyi Region, Shan (North) and Ayeyarwaddy Region.

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KEY FINDINGS — MYANMAR HEALTHCARE ACCESSIBILITY (2020)
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FACILITIES

Total facilities analyzed:	1,130
With bed capacity data:	1,091 (97%)
Total beds recorded nationally:	53,975

REGIONAL HIGHLIGHTS

State/Region with MOST facilities:	Sagaing (141 facilities)
State/Region with FEWEST:	Kayah (17 facilities)

ACCESSIBILITY INDEX

Most accessible township:	Latha (Yangon) — score: 1.000 [Very High]
Least accessible township:	Mongla (Shan (East)) — score: 0.000 [Very Low]
Townships with 'Very Low' tier:	66 (20% of all townships)

SPATIAL COVERAGE

Townships with local facility:	98.5%
Average nearest facility distance:	9.2 km
Townships > 20 km from any facility:	23

Key Findings

Among the total facilities 1,130, 68.1% is the station hospitals, 20% as township hospitals and 6% as general hospitals. Among 53,975 beds recorded, Sagaing Ayeyarwady, Mandalay have the highest facilities count, while Kayah and Nay Pyi Taw have the lowest, following the trend of the geographical size. As for the bed capacity, the highest bed capacity regions like Yangon (Latha Township) and Nay Pyi Taw (Oke Ta Ra Thi Ri Township) has coverage of 250 beds per 10,000 population, while the underserved areas like Ayeyarwaday and Sagaing have low bed per population, pointing that the current facilities in those regions need more capacity and resources, despite having high number of facilities.

As a gap between distance to the nearest hospital, Kayin and Bago (East) have the highest person-km travel burden to get to the nearest health facility. The average nearest facility distance is 9.2km and there are 23 townships which are less than 20km distance from the health facility.

As for the township level, Shan (North) has the most remote townships (for e.g - Narphan, Pangwaun), where the distance to nearest hospital is more than 60km. Notably, only 28% has a regional level area within a 5km buffer of public hospitals. Based on the healthcare accessibility index, 20% of all townships in Myanmar have a very-low level index. According to the index, Yangon and Nay Pyi Taw are the best-served areas while Shan (East) and Shan (North) have significant challenges with low index.

Conclusion

The report analyses detail on the geographical area of Myanmar in different levels with bed capacity, facility count, distance to the nearest hospital, bed per population and healthcare accessibility index, healthcare risk exposure. The findings give critical information in understanding the general overview of Myanmar's public healthcare

facility and support in allocation of resources of public healthcare infrastructure and resources from a multidisciplinary perspective.

Limitation

The dataset is between the 2016 census population and 2020 health facilities record which may change in population and facilities count in 2026. The facilities mainly focus on public hospitals recorded, and private hospitals are not included. Moreover, the scope is only on bed capacity, number of stations, township and general hospitals without considering other factors such as number of public health staff and quality of the medical equipment. Due to time limitation, in calculating the nearest facility, the Euclidean distance is used as straightline in this report which has limitations on the diverse topography and road networks, which can be further explored with network distance or other relevant methods.

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